

Probability - An Introduction

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Some definitions:

- **Probability** - the chance some event will occur.
- ✓ ▪ **Permutations** - all the ways an event can occur when order matters.
E.g., tossing a coin three times and get HTH in that order.
- ✓ ▪ **Combinations** - all the ways an event can occur when order does not matter.
E.g., tossing a coin three times and get 2 heads and 1 tail in any order (HHT, THH, HTH).

Counting Problems

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Example The Parkers are taking a photograph of their four children—Alan, Bianca, Callie, and Dennis. They plan to stand in a line. How many different ways can the children line up?

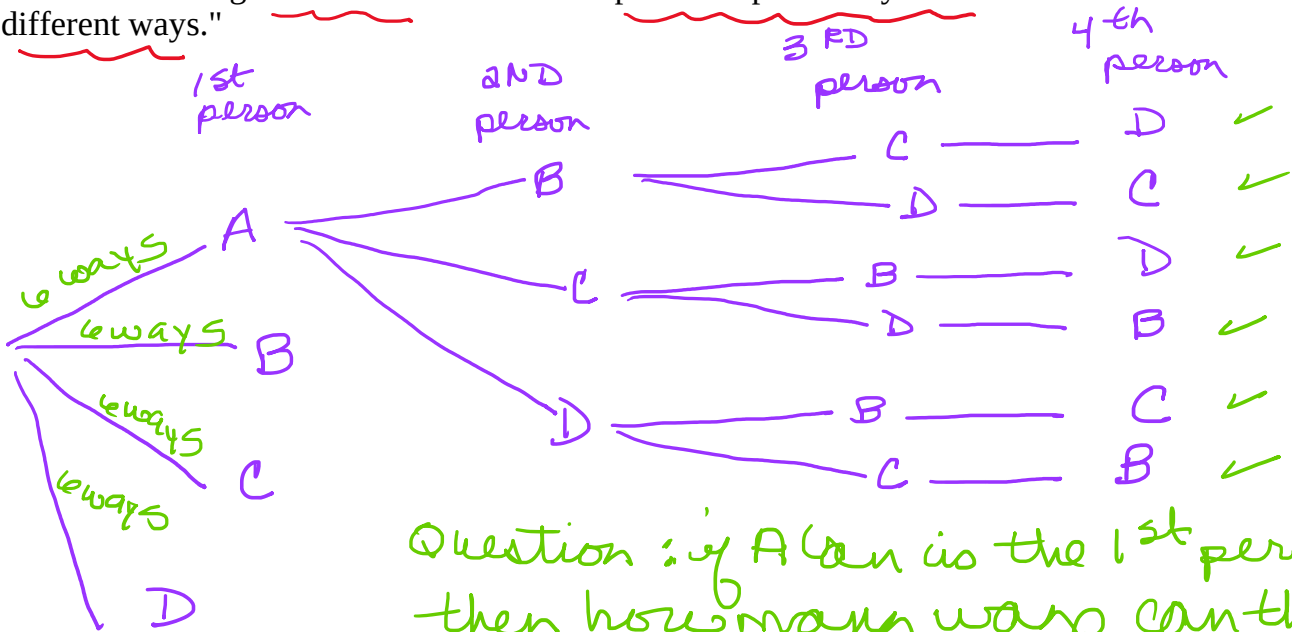
We will solve this problem using a **Probability Tree**.

Let A = Alan, B = Bianca, C = Callie, and D = Dennis.

ORDER MATTERS
→ Permutation

Probability Trees can be draw horizontally or vertically.

We will be finding Permutations because the problem specifically states "different ways."



Question: if Alan is the 1st person, then how many ways can the other 3 line up? 6 ways

How many total ways can these 4 people line up? 24

Example Emily wants to wear red, white, and blue for a holiday. She can choose from two pairs of sneakers—blue or white. She has three shirts—red, white, or blue—and three pairs of shorts—red, white, or blue. She wants to wear one red item, one white item, and one blue item. How many outfits can she make?

sneakers
1st choice

shirts
2nd choice

shorts
3rd choice

blue sneakers

red shirt
white

white ①

red ②

~~blue~~

4 outfits

white sneakers

~~blue~~

blue

3

R

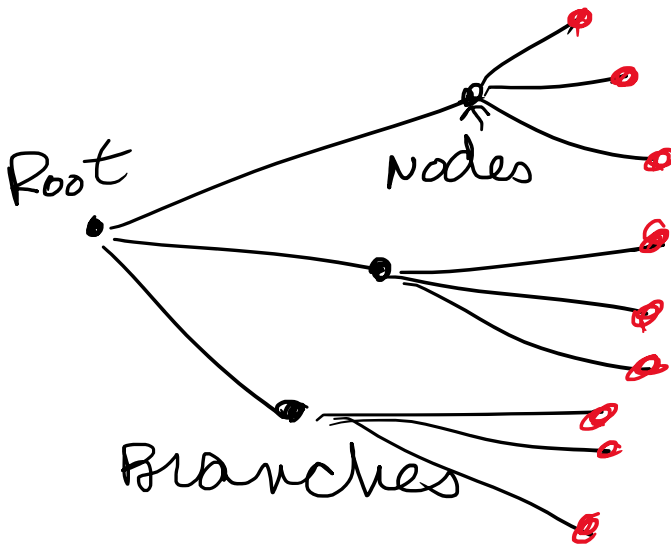
~~blue~~

B

Red

4

A TREE



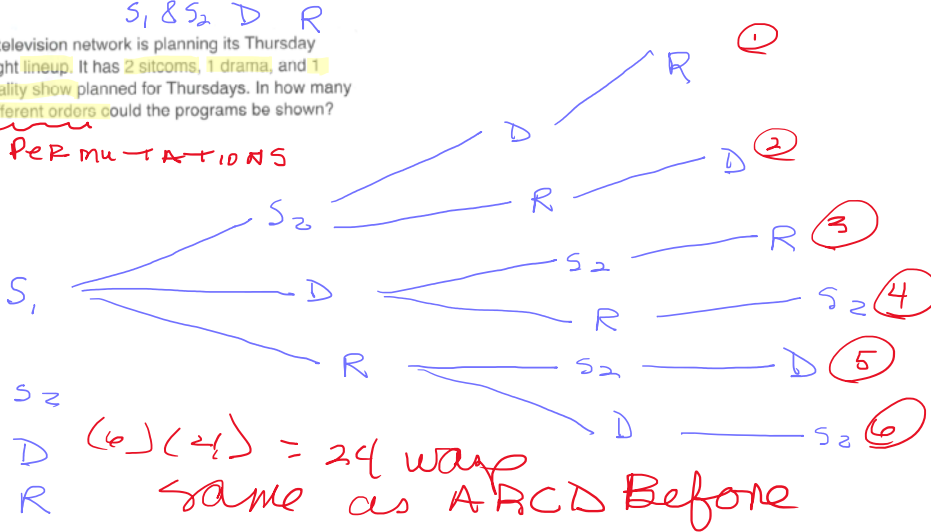
COUNT
WAYS

Examples

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1. A television network is planning its Thursday night lineup. It has 2 sitcoms, 1 drama, and 1 reality show planned for Thursdays. In how many different orders could the programs be shown?

PERMUTATIONS



2. A car can seat 2 people in the back and 2 in the front. A mother and her 2 children get in the car. In how many different ways could the family be seated if the mother always has to be the driver and either child can sit up front?

Let C_1 = 1st child
 C_2 = 2nd child

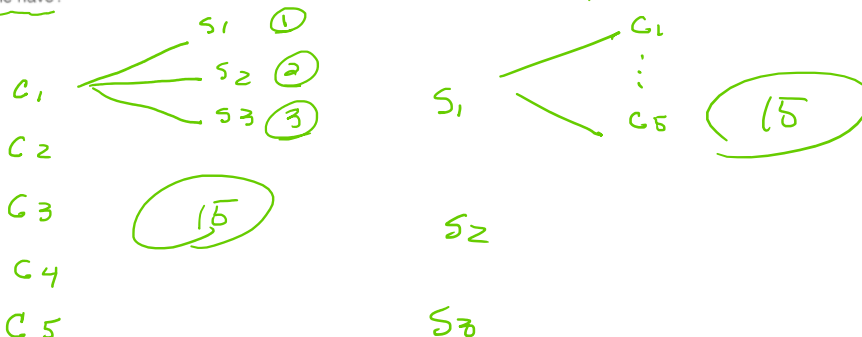
passenger seat 1 Back left seat 2 Back right seat 3

C_1	C_2	—	✓
C_2	—	C_1	✓
C_1	C_2	—	✓
C_2	—	C_1	✓
—	C_1	C_2	✓
—	C_2	C_1	✓

6 ways

3. Jan is ordering custom shirts for her business. There are 5 colors and 3 styles of shirts to choose from. How many different choices does she have?

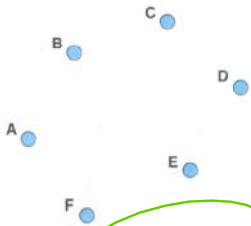
Note:



4. In the figure, how many different line segments

1 $C(6, 2)$

4. In the figure, how many different line segments can be drawn that connect exactly two points?
(Note: AB is the same as BA.)



15 ways

NOT A PERMUTATION; A COMBINATION!

$n = 6$ items

selecting $r = 2$ items

AB

AC

BC

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

AD

BD

CD

AE

BE

CE

DE

AF

BF

CF

DF

EF

(5)

(4)

(3)

(2)

(1)

$$C(6, 2)$$

$$= \frac{6!}{4! \cdot 2!}$$

$$= \frac{6 \times 5 \times 4!}{4! \cdot 2!}$$

$$= 30/2$$

$$= 15 \text{ ways}$$

More Examples

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5. The directions for a game say that a player must take two different actions on his or her turn.

The player's choices are:

5 - choices

- 1. Draw a card
- 2. Play a card
- 3. Take a move action
- 4. Capture an opponent
- 5. Buy an item from the bank

The order that the player takes the actions matters. Moving and then playing a card is different from playing a card and then moving.

How many different two-action turns are possible?

A PERMUTATION

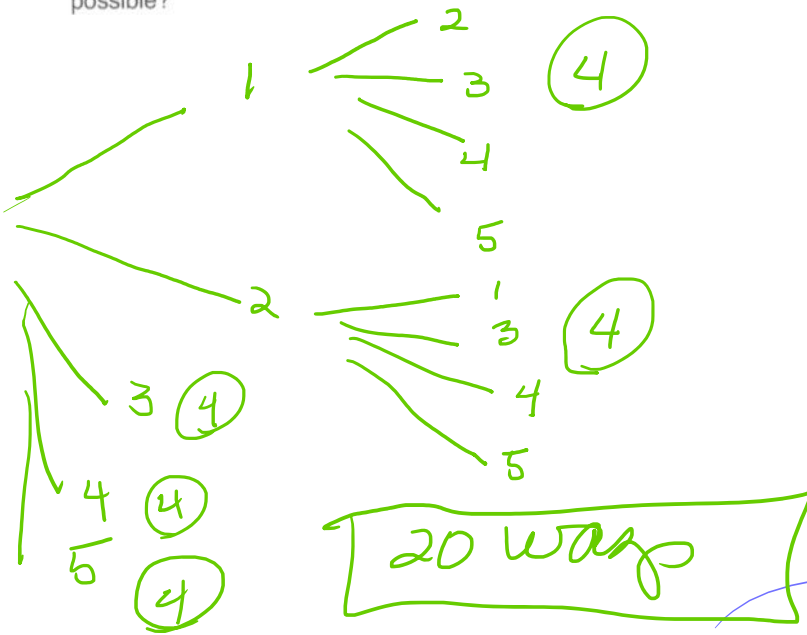
5 choose 2

$$P(n, r) = \frac{n!}{(n-r)!}$$

$$P(5, 2) = \frac{5!}{3!}$$

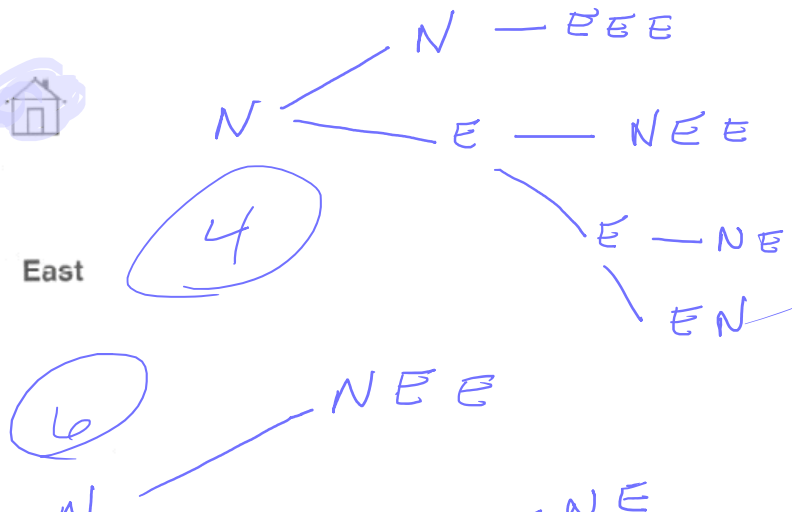
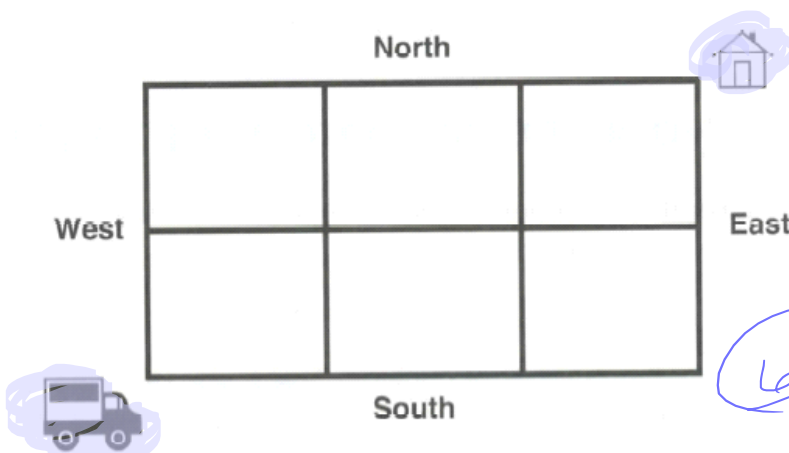
$$= \frac{5 \times 4 \times \cancel{3!}}{\cancel{3!}}$$

$$= 20$$



6. The truck in the drawing needs to make a delivery to the house in the top right corner. If the truck must stay on the lines and only travel north or east, how many different routes can the truck take to get to the house?

10 ways





South

